

Solution Summary

The HDI Prediction System is implemented as a Flask web application that loads a pre-trained Linear Regression model and returns a predicted Human Development Index value based on four user-provided indicators.

Core Implementation Steps

- Load and clean the 2021 HDI dataset covering approximately 195 countries.
- Select four features: Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita.
- Train a Linear Regression model using scikit-learn and evaluate with R^2 (approx. 0.98).
- Serialize the trained model using pickle for use in the Flask application.
- Build Flask routes: one to render the input form, one to handle form submission and prediction.
- Render the predicted HDI value back to the user through the dark-themed result page.

Key Code Components

- `model = pickle.load(open('hdi_model.pkl','rb'))` — loads the trained model at startup.
- `@app.route('/predict', methods=['POST'])` — handles form submission and returns the prediction.
- `request.form.get(...)` — retrieves each of the four indicator values from the form.
- `model.predict([[...]])` — generates the HDI prediction from the input features.

Challenges and Fixes

- Column name mismatches between the form fields and dataset — resolved by aligning names exactly.
- Windows PATH issues with python/pip — resolved by using the 'py' launcher consistently.
- Incorrect model file path on deployment — fixed by using relative paths from the app root.

Outcome

The final solution is a working, publicly deployed Flask application that accepts four development indicators and instantly returns a predicted HDI score, meeting the project's core objective.